

Math 70

Week 6

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1 Problems

Example 1.1. You throw a fair coin consequently. What is the probability that HHH appears before TT?

Example 1.2. In an urn there are 6 green, 6 yellow, 6 red, and 6 blue balls.

1. You pick two balls. What is the probability that they are the same color.
2. You pick three colors. What is the probability that you exactly all colors are different?
3. You pick three colors. What is the probability that you exactly two colors are the same? (Eg BBR, or RRY).

Example 1.3. You throw a dice repeatedly. Let X be the number of throws until you get three 6s. What is the expected number of throws?

Example 1.4. Let $X \sim \text{Poi}(\lambda)$. Compute $\text{Var}(X)$.